

m10ASYS ASCOM Driver

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Introduction

ASCOM dome/roof driver for observatories using the ELK M1 Gold and m1OASYS architecture.

Overview

The m1OASYS ASCOM Roof Driver provides observatory roof automation through direct ELK M1/XEP integration using the Elk ASCII protocol. The driver supports roll-off roof control, telemetry, calibration, watchdog protection, and optional mount-safe interlocks.

Supported Applications

- NINA
- Voyager
- Sequence Generator Pro (SGP)
- CCD Commander
- Custom ASCOM automation software

No warranty as to the operation or automation of the observatory equipment; it simply provides components for 'Do It Yourself' automation projects.

Caution! Automation of any rolling roof observatory requires special attention with regard to the location of individuals in the observatory, as well as the position of the telescope/mount, prior to moving the roof to ensure no one or nothing obstructs the roofs path. Serious injury to individuals and/or damage to telescope, mount or the observatory may occur if this advice is not followed. We recommend that all users of m1 ASCOM driverLive implement safety policies and procedures regarding the operation of their rolling roof. In addition we strongly recommend installation of safety sensors that will prevent or stop movement of the roof should individuals or equipment move into an unsafe position before or during roof movement. For remote/automated operation of rolling roofs it is recommended that the *"Allow Remote Apps to Open/Close the Roof"* check-box in the m1 ASCOM driverLive Live software be unchecked if the roof does not safely clear the scopes/mounts. Misuse or careless operation can cause serious damage to your equipment or individuals. The author of the software and the distributor of the equipment (1) make no warranty or representation to the user as to the operation of this System and (2) is NOT responsible for any injury or damage that may occur from careless implementation, operation or misuse. **Opening or Closing a roll-off roof is not a trivial task and the operator needs to take all necessary precautions before starting the roof motor. Damage to equipment or persons can occur when the roof is in motion so all necessary precautions should be taken before executing such a command.**

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System Requirements

Hardware Requirements

- Windows PC
- ASCOM Platform
- ELK M1 Gold with XEP Ethernet interface

- Configured ElkRP automation rules
- Optional Hall-effect pulse sensor
- Optional scope-safe sensors
-

Driver Architecture

Component	Responsibility
ELK M1	Hardware authority and sensor evaluation
ASCOM Driver	Roof control, telemetry, safety enforcement
Client Software	Observatory sequencing and telescope control

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Elk m1 Rules, Texts, Counters, Etc.

ELK M1 Configuration

Overview

The m1OASYS ASCOM Roof Driver communicates directly with the ELK M1/XEP using Elk ASCII text commands over TCP/IP.

The ELK provides:

- Hardware relay control
- Roof position sensing
- Optional Hall pulse telemetry
- Optional scope-safe interlock evaluation
- Status responses back to the ASCOM driver

The ASCOM driver is responsible for:

- Roof automation
- Telemetry interpretation
- Safety enforcement
- Calibration
- Watchdog protection
- ASCOM interoperability

Required ELK Configuration

Required Tasks

The following ELK tasks are expected by the driver.

	Task	Purpose
Task 1		Open Roof
Task 2		Close Roof

Task 3

Stop Roof

Example names:

Task Number	Suggested Name
1	Open Roof
2	Close Roof
3	Stop Roof

Required Zones

Roof Open Limit

Zone	Purpose
Zone 11	Roof Opened Sensor

Expected behavior:

- NOT SECURE = Roof fully open

Roof Closed Limit

Zone	Purpose
Zone 12	Roof Closed Sensor

Expected behavior:

- SECURE = Roof fully closed

Optional Zones

Scope Safety Sensor

Zone	Purpose
Zone 7	Scope Safe Sensor

Expected behavior:

- SECURE = Telescope safe for roof motion
- NOT SECURE = Unsafe for roof motion

This zone is only required if:

- "Enable Mount Safety Sensors" is enabled in the ASCOM Setup Dialog.

Hall Pulse Sensor

Zone	Purpose
Zone 5	Roof Motion Pulse Sensor

Used for:

- pulse counting
- calibration
- percent-open tracking
- watchdog protection

This zone is optional.

Required Counters

Roof Pulse Counter

Counter	Purpose
Counter 7	Roof motion pulse count
The driver polls this counter during motion when pulse telemetry is enabled.	

Required Text Definitions

Open Roof Command

Text Number	Value
2	09tn00100C4^M

Close Roof Command

Text Number	Value
3	09tn00200C3^M

Roof Status Query

Text Number	Value
7	09xx00100B6^M

Scope Safety Query

Text Number	Value
20	11xx005sensoron0042^M

Pulse Counter Query

Text Number	Value
25	07cv00729^M

Required Driver Responses

Roof Open Response

ELK sends:

0DXX001open0039^M

Driver interprets:

- roof fully open

Roof Closed Response

ELK sends:

0FXX001closed006F^M

Driver interprets:

- roof fully closed
-

Roof Unknown Response

ELK sends:

10XX001unknown00EE^M

Driver interprets:

- roof moving / intermediate state
-

Scope Safety Responses

Scope Safe

ELK sends:

0FXX002Secure0081^M

Driver interprets:

- roof motion allowed
-

Scope Unsafe

ELK sends:

12XX002NotSecure0063^M

Driver interprets:

- roof motion blocked
-

Required Automation Rules

Open Roof Task

WHENEVER Open Roof (Task 1) IS ACTIVATED

THEN TURN Open Roof Relay ON

Close Roof Task

WHENEVER Close Roof (Task 2) IS ACTIVATED

THEN TURN Close Roof Relay ON

Stop Roof Task

WHENEVER Stop Roof (Task 3) IS ACTIVATED

THEN TURN Open Roof Relay OFF

THEN TURN Close Roof Relay OFF

Roof Status Query Rules

The driver periodically requests roof status using:

09xx00100B6^M

ELK must respond with:

- OPEN
- CLOSED
- or UNKNOWN

based on current limit switch state.

Pulse Counting Rule

WHENEVER Roof Pulse Zone BECOMES NOT SECURE

THEN ADD 1 TO Counter 7

Pulse Counter Reset Rules

WHENEVER Open Roof Task ACTIVATES

THEN SET Counter 7 TO 0

WHENEVER Roof Closed Sensor BECOMES SECURE

THEN SET Counter 7 TO 0

Scope Safety Rules

WHENEVER THE FOLLOWING TEXT IS RECEIVED:

11xx005sensoron0042^M

AND Scope Safe Zone IS SECURE

THEN SEND:

0FXX002Secure0081^M

WHENEVER THE FOLLOWING TEXT IS RECEIVED:

11xx005sensoron0042^M

AND Scope Safe Zone IS NOT SECURE

THEN SEND:

12XX002NotSecure0063^M

Notes

CRC Framing

The driver internally generates CRC-framed Elk packets automatically.

ELK rules must match the fully framed transmitted text including:

- packet length
 - CRC
 - carriage return (^M)
-

Optional Features

The following driver features are optional:

Feature	Required Hardware
Pulse Telemetry	Hall sensor
Percent Open Tracking	Hall sensor
Scope Safety Interlock	Scope-safe sensor

The driver supports traditional limit-switch-only operation when optional hardware is not installed.

Recommended Architecture

ELK Responsibilities

- Relay control
- Sensor authority
- Pulse counting
- Safety state determination

ASCOM Driver Responsibilities

- Roof automation
- Telemetry processing
- Calibration
- Watchdog protection
- Safety enforcement

Client Software Responsibilities

- Observatory sequencing
- Telescope parking
- Imaging automation

This separation provides a reliable and maintainable observatory control system.

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Optional

Installation

m1 ASCOM driver is available via Internet download.

Important: m1 ASCOM driver requires the installation of the ASCOM Platform, available at <http://ascom-standards.org>.

Internet Installation

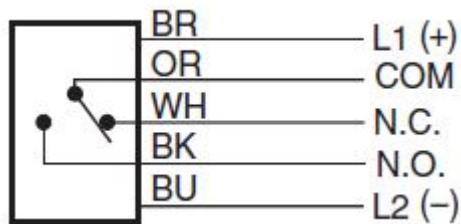
1. Download the latest m1 ASCOM driver driver and software.
2. You will be prompted to either 'Run' the installer or you can save it to a folder; double clicked the saved file with start the installer.
3. Follow the on screen instructions to complete the installation.

- Once the software has successfully installed you must enter the setup areas to input the proper setting before attempting to connect your hardware.

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Scope Position Sensor

Eaton 1451E



Supply power to the sensor via the m1 ASCOM driver board's +12v (brown) and -12v (Blue), then connected the common (Orange) and normally open (N.O. Black) sensor wires to the appropriate zone input.

Create rules to ensure the sensor is secure before moving the roof, and you can even wire the power through a m1 ASCOM driver output relays to give you control of turning the IR beam on and off before and after opening and closing commands are sent.

Rule examples:

WHENEVER THE FOLLOWING TEXT IS RECEIVED: "11xx005sensoron0042^M" THROUGH PORT 0
THEN TURN Scope IR Sen (Out 13) ON

WHENEVER Open Roof (Task 1) IS ACTIVATED
AND Scope 1 Sensor 1 (Zn 7) IS SECURE
THEN TURN Open Roof (Out 9) ON FOR 5 SECS

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Setup

Setup Configuration

1. Install the ASCOM Platform.
2. Install and register the m1OASYS ASCOM Roof Driver.
3. Configure the ELK M1 XEP IP address and port in the Setup Dialog.
4. Enable optional pulse telemetry if a Hall sensor is installed.
5. Enable optional scope safety interlock if safety sensors are installed.
6. Verify ElkRP text and automation rules are configured properly.
7. Connect the driver from NINA or another ASCOM client.

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Hall Sensor Calibration

Calibration

The driver supports automatic roof calibration using Hall-effect pulse counting. During calibration, the roof opens fully while the driver records the pulse count. The learned value is saved automatically and used for percent-open calculations.

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Features

Features

- ASCOM Dome interface support
- Open / Close / Abort roof control
- Live shutter state telemetry
- ELK M1 TCP/IP integration
- Optional Hall-effect pulse telemetry
- Automatic roof calibration
- Percent-open roof tracking
- Movement timeout watchdog
- Pulse watchdog protection
- Optional scope-safe interlock support
- Persistent calibration storage
- Integrated live telemetry status window

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Supported Software

Supported Applications

- NINA
- Voyager
- Sequence Generator Pro (SGP)
- CCD Commander
- Other ASCOM automation software

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Hardware Required

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Telemetry

- Roof state
- Percent open
- Pulse count
- Fault state
- Reconnect status
- Watchdog events
- Calibration status

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Safety Features

Safety Features

Movement Timeout Watchdog

Detects stalled motion, failed relays, missing limit transitions, or roof jams. Automatically aborts motion and reports a fault state.

Pulse Watchdog Protection

When pulse telemetry is enabled, the driver monitors Hall pulses during roof movement. If motion pulses stop unexpectedly, the driver enters a fault state.

Scope Safety Interlock

Optional hardware-based scope-safe interlocks can be enabled through ELK rules. When enabled, the driver blocks roof movement unless the ELK reports the telescope is in a safe position.

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